MEMORY BOARD
CORE BOARD
PERIPHERAL BOARD
J1
USB-C 5V POWER BREAKOUT
D1
1N5819
PWR_FLAG
J3
EXTERNAL 5V
U2
AS6C1008-55PCN
Z84C0020PEC
A1
Raspberry Pi Pico 2 W
J2
PICO GPIO EXPANSION
A2
0.91in SSD1315 128X32 I2C OLED
U3
SN74HCT32N
U9
SN74AHCT245N Pico-to-bus
U10
SN74LVC245AN bus-to-Pico
RN1
RN2
RN3
RN4
RN5
RN6
8x10k bussed
4x10k bussed
BOOT_CONTROL
ADDRESS
DATA
HCT_TO_CPU
PICO_DATA
PICO_TO_BUFFER
CPU_TO_BUFFERS
BUFFERED_CONTROLS
HCT_TO_SRAM
CPU_MONITOR
PICO_MONITOR
PORT_MONITOR
DATA_CONTROL
RESET_N
C1
C2
C3
C4
C5
C6
C7
C8
C9
C10
C11
100n
22u
100u

Z80 is the sole SRAM address/write master. Pico injects instructions with SRAM reads inhibited. RESET_N directly joins Pico and Z80 at 3.3V.

J1: onboard USB-C 5V breakout, not a regulator.
21.6 x 12.7 mm body; photo-scaled outline, dry-fit first.
Solder ALL six 2.54mm pins into 1.0mm carrier holes.
Photo pad order: CC2, D+, D-, CC1, GND, VBUS.
Only VBUS -> +5V and GND wired; other pads isolated.
No module screw holes; insulate/support underside.
Use regulated 5V; verify both CC sink resistors.
J3 is auxiliary: only ONE supply at a time.
Keep D1 to Pico VSYS; never join Pico VBUS.

J2: 3.3V expansion/probe header, not a 5V input.
1: GP22/SDA; 2: GP26/SCL, shared with A2.
3: Pico 3V3 output; 4: GND.
GP8 is dedicated to SRAM bank select.

BANK SELECT: Pico physical pin 11 / GP8 -> U4.17.
RN6.5: 10k to RN6.1 / GND (breadboard R24 equivalent).
U4.3 -> U2.2 / A16 (5V side); LOW selects bank 0.

A2: socketed Stage 10 0.91in SSD1315 128x32 status OLED.
Module 38.00 x 12.00 mm; 4-pin 2.54 mm header.
Header centres 1.50mm from left; SDA 2.19mm from top.
Header order GND, VCC, SCK, SDA bottom-to-top.
SDA = J2.1 / GP22; SCK = J2.2 / GP26.
VCC = J2.3 / 3V3; GND = J2.4 / GND.
I2C pull-ups to 3.3V only; address 0x3C.
Verify socket height, pin fit and insulated module support.

RN5: four 3V3 startup pull-ups (breadboard R17–R20). RN6: four startup pull-downs (breadboard R21–R24). Both data paths default disabled.

Native KiCad 10 schematic generated from the verified README pin map
Z80ROMlessSBC
Sheet: /
File: z80_romless_sbc.kicad_sch
Title: Z80ROMlessSBC – Assisted Loader and Four-Layer PCB
Size: A0
Date: 2026-11-02
Rev: 3
KiCad E.D.A. 10.0.7
Id: 1/1